

QRAFT TECHNOLOGIES, LONG-TERM EQUITY TEAM, 2022 – 2024

A Statistical Multi-Factor Alpha Pipeline

How 360 candidate factors become one Composite Alpha Score and a monthly long/short US equity portfolio for the SoftBank fund.

Juyeong Lee, AI Quant Researcher

How do you build a profitable multi-factor portfolio?

Three questions the pipeline had to answer, in order



Generate many factors

Build a library of 360 candidate factors from fundamentals, prices and sentiment — value, growth, quality, low volatility, size, momentum.



Screen them statistically

Keep only factors whose decile spread, Fama-French alphas and monotonicity are all significant over a rolling 15-year window, then strip redundancy.



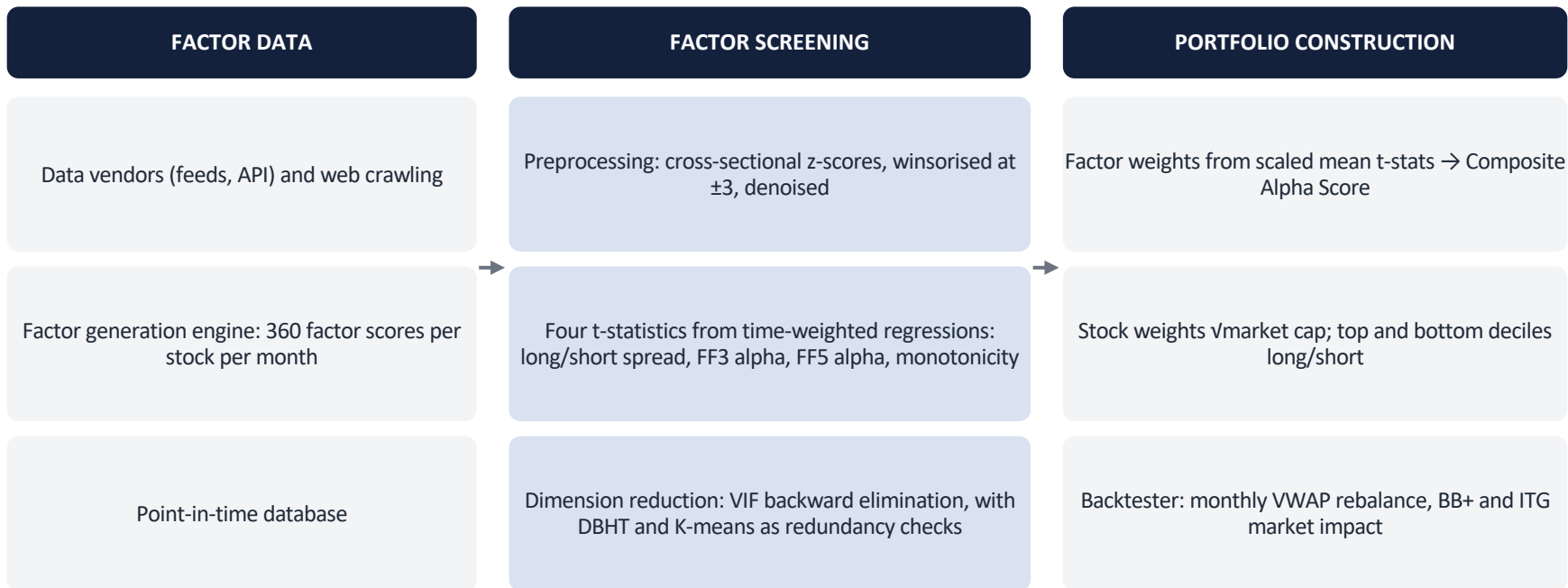
Combine into one score

Weight the survivors by statistical conviction into a Composite Alpha Score, rank 1,500 stocks and hold the top and bottom deciles.

Rules-based, monthly, and testable out of sample: every step re-runs at each month-end on data available at the time.

End-to-end architecture

Factor data → factor screening → portfolio construction and backtest



Each column runs at month-end; the portfolio is applied on the first trading day of the next month.

Universe and base settings

A liquid 1,500-stock universe and a handful of fixed rules



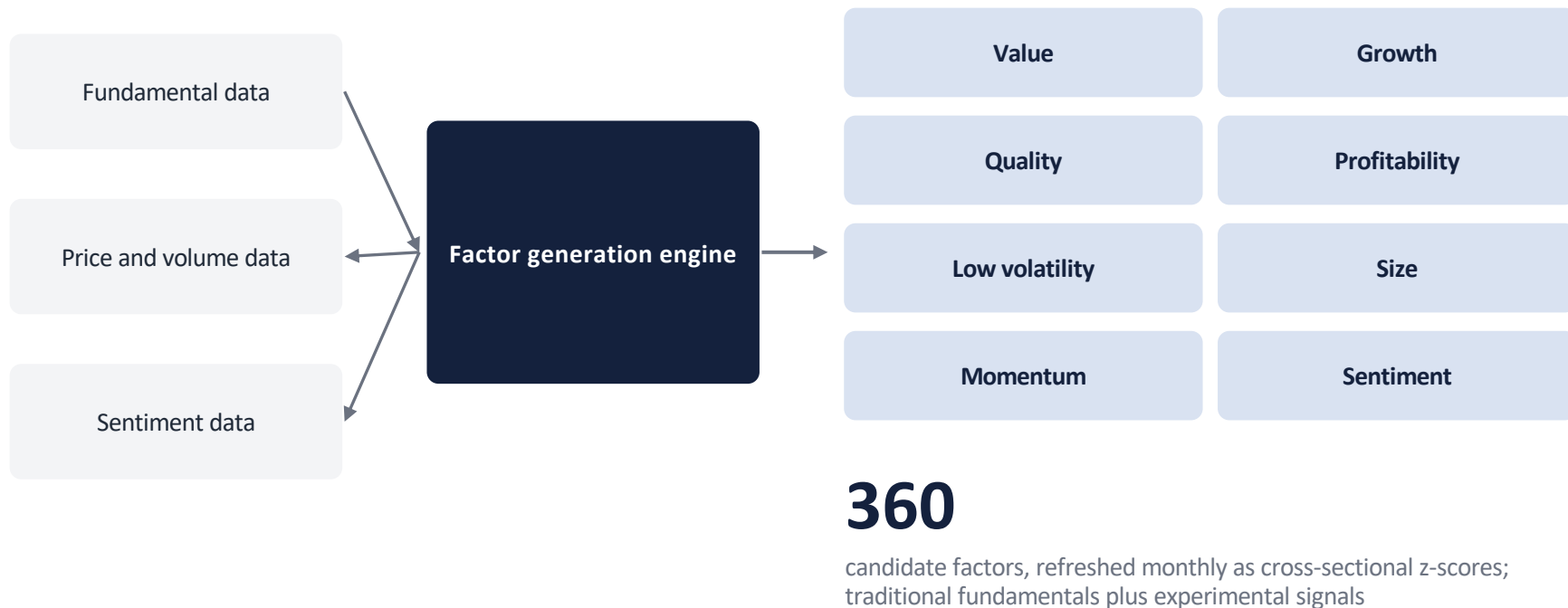
Liquid enough for scalable weighting schemes; few hard-to-borrow names.

Base settings

- Monthly: run the algorithm at month-end, apply the portfolio on day 1 of the next month
- Execution modelled as all-day VWAP — low impact, rebates as liquidity provider
- Factors converted to cross-sectional z-scores, winsorised at ± 3 (about 5 outliers a month per 1,500 stocks)
- Stock weights: square root of market cap — between equal and cap weighting
- Deciles: top 150 long, bottom 150 short

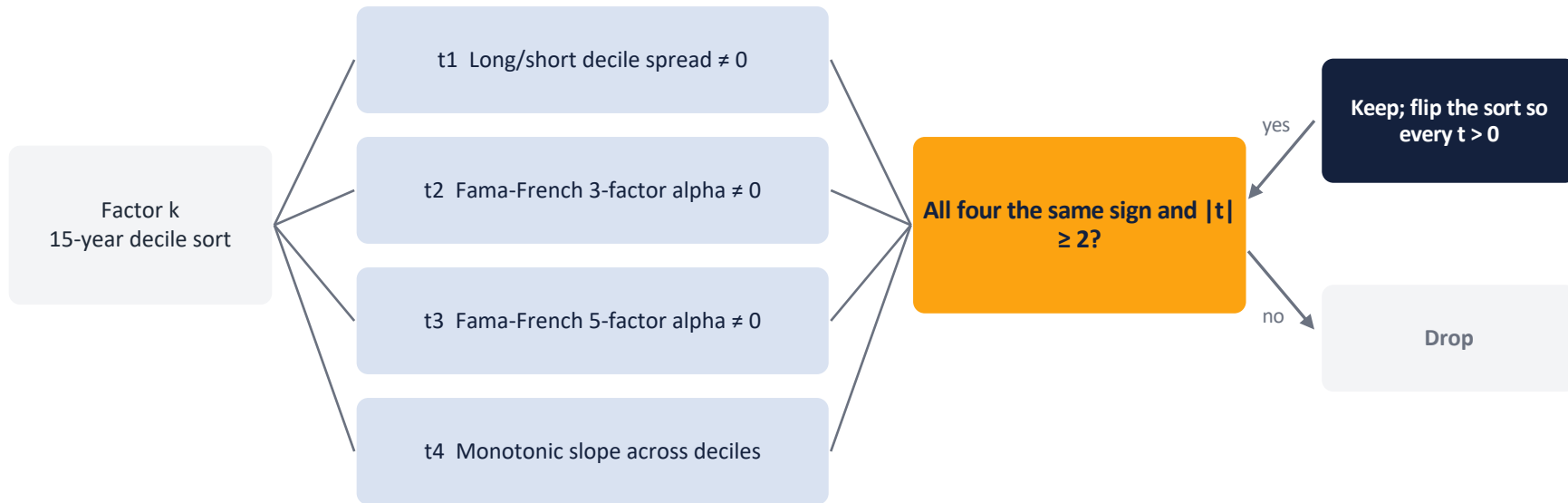
A library of 360 candidate factors

Three data families feed one generation engine



Screening: four t-statistics per factor, every month

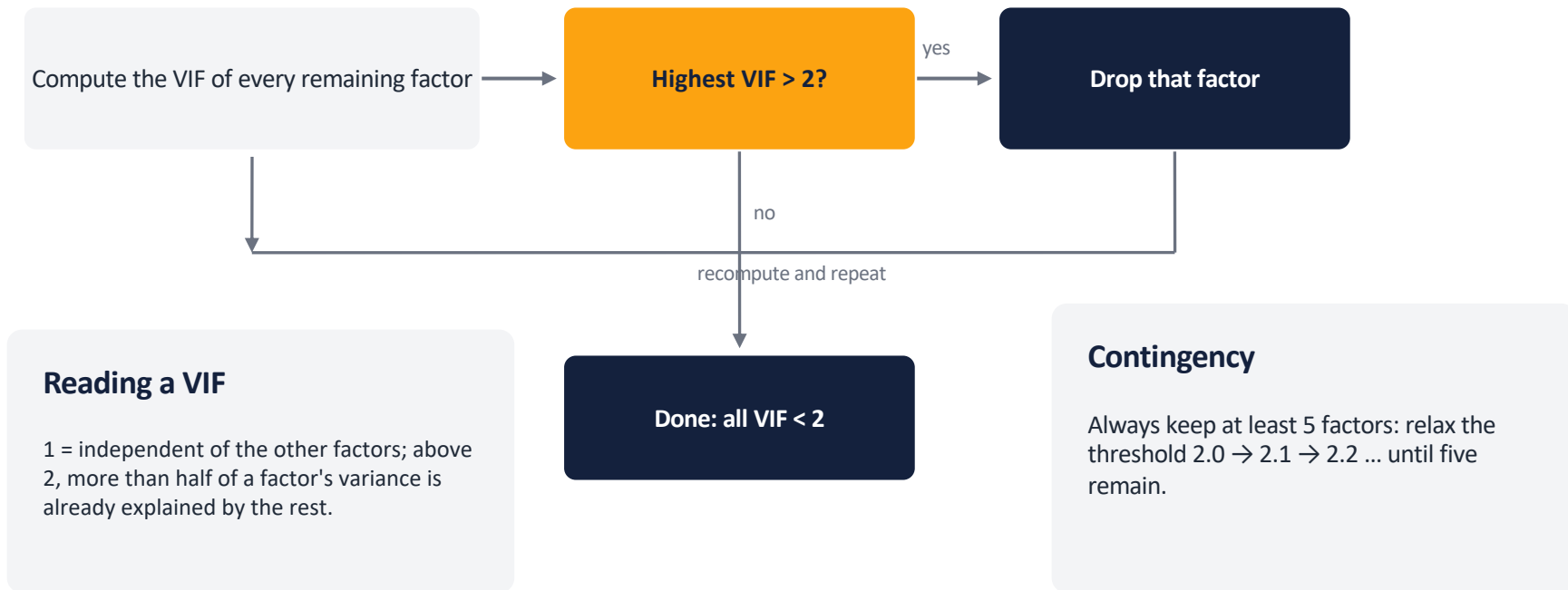
Rolling 15-year (180-month) time-weighted regressions on decile portfolios



- Time weights $W = \exp(-\text{elapsed} / 179)$: the latest month counts 1.0, the oldest about 0.37 — a full business cycle, tilted to recent data; the window rolls monthly, so the backtest is out of sample from 2002.
- Contingency when fewer than 5 factors survive: relax the monotonic threshold $2 \rightarrow 1.5 \rightarrow 1 \rightarrow 0.5 \rightarrow 0$, then t1–t3 to 1.96 and 1.90.

Dimension reduction: VIF backward elimination

Remove redundant factors one at a time until every survivor has a variance inflation factor below 2



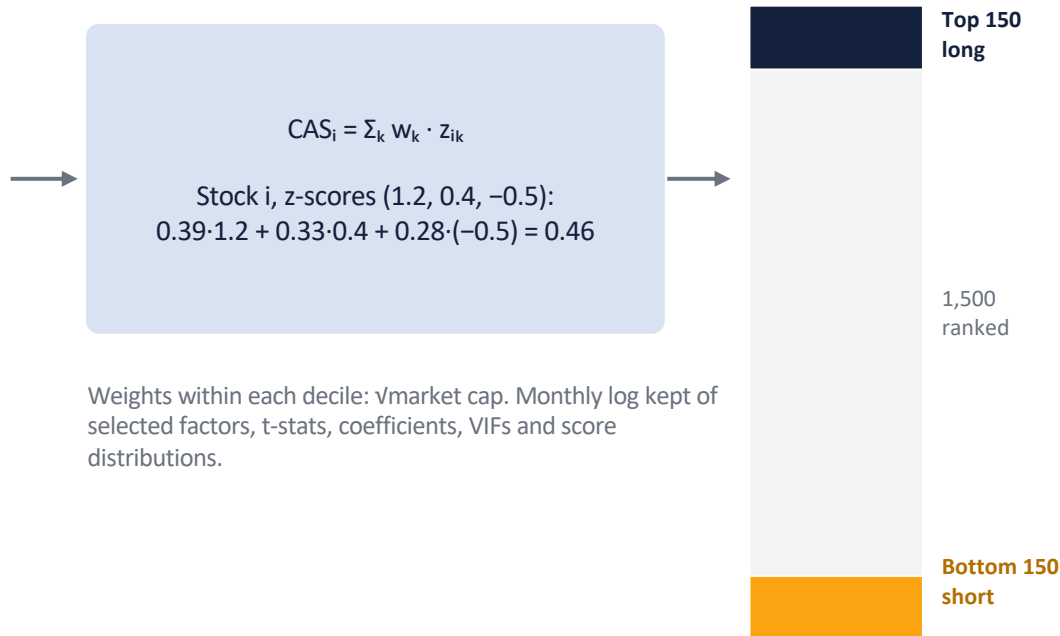
DBHT and K-means clustering of factor correlations were run alongside as a check on redundant factor families.

Composite Alpha Score: one number per stock

Surviving factors are weighted by statistical conviction, then every stock is ranked

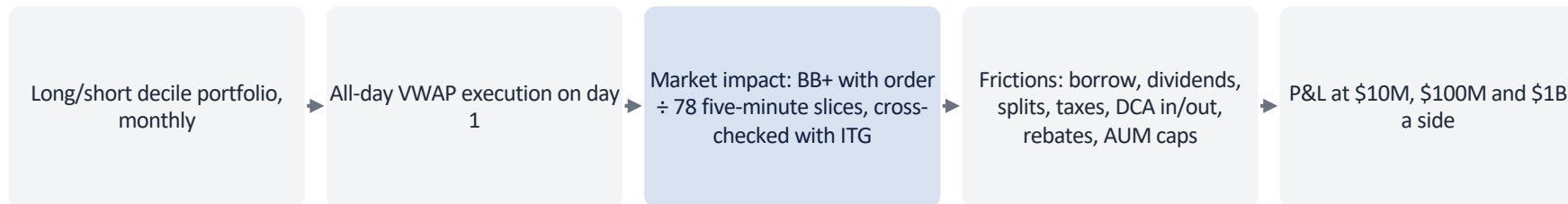
Factor (illustrative)	Mean t	Weight
Factor A	3.1	0.39
Factor B	2.6	0.33
Factor C	2.2	0.28
Sum		1.00

Weight = mean of the four t-stats, scaled to sum to one, so higher-conviction factors count more.

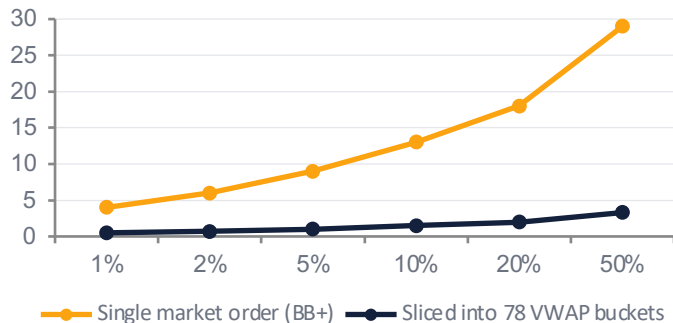


A backtest that mirrors live trading

Market impact and every friction the fund would actually pay



Impact cost (bp) vs order size (% of ADV), illustrative shape



93%

agreement with realized P&L; used for the fund's strategy backtests

Outcome

Consistent excess return over SPY for the SoftBank fund (figures confidential). Tracked: long, short and combined returns and P&L, volatility, Sharpe, % positive days, max drawdown, Sortino and seasonality; optimised variant via the Barra optimiser.

What it delivered

A 360-factor library reduced every month to a small, orthogonal, statistically significant set — by rules, not judgement.

One Composite Alpha Score per stock, weighted by conviction, driving a 150-long / 150-short decile book.

A backtester that reproduces realized P&L within 93%, so simulated alpha survives execution.

Excess return over SPY for the SoftBank fund and a scalable platform for further factor research.

Python, NumPy, Numba, pandas, Snowflake, SQL, Redis

Juyeong Lee, juyeong.lee0839@gmail.com